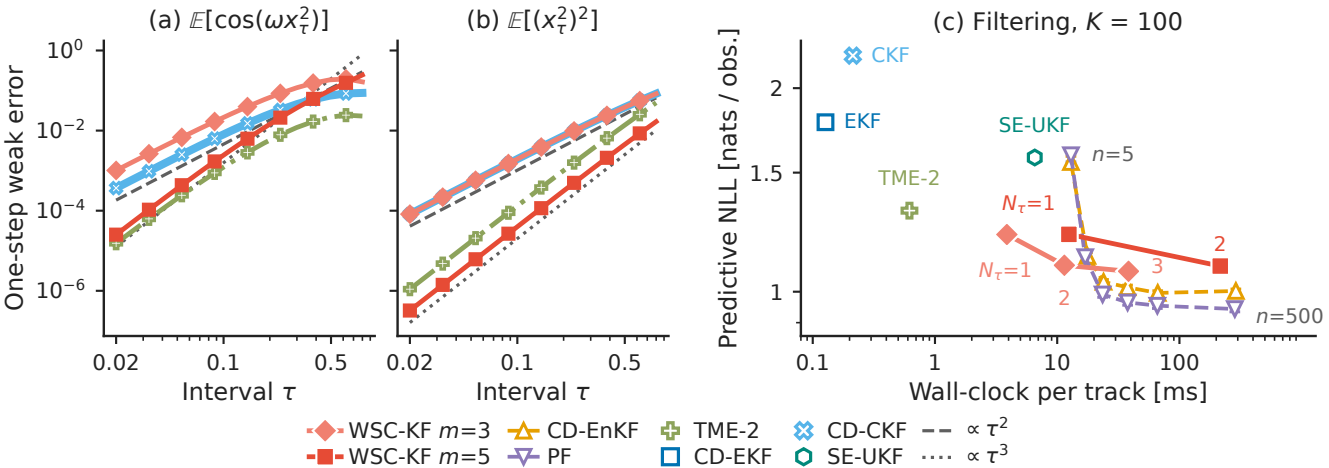
(a) $\mathbb{E}[\cos(\omega x_\tau^2)]$
(b) $\mathbb{E}[(x_\tau^2)^2]$
(c) Filtering, $K = 100$
One-step weak error
$10^0$
$10^{-2}$
$10^{-4}$
$10^{-6}$
0.02
0.1
0.5
Interval $\tau$
Predictive NLL [nats / obs.]
2
1.5
1
0.1
1
10
100
Wall-clock per track [ms]
CKF
EKF
SE-UKF
TME-2
$n$=5
$N_\tau$=1
$N_\tau$=1
2
3
2
$n$=500
WSC-KF $m$=3
WSC-KF $m$=5
CD-EnKF
PF
TME-2
CD-EKF
CD-CKF
SE-UKF
$\propto \tau^2$
$\propto \tau^3$